

Lab 1: Requirements Engineering & UML Use-Case Modelling

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Section: I

System Name: Academic Elective Bidding & Allocation System

Use-Case Flow Specification

- **Use Case ID:** UC-01
- **Use Case Name:** Submit Elective Bids
- **Primary Actor:** Student
- **Secondary / Supporting Actors:** Academic Registrar

Precondition:

1. The student is authenticated and logged into the university portal.
2. The elective bidding window is opened.
3. The student has 100 bidding credits available.

Postcondition:

1. The list of electives bid for, and credits allocated is stored in the database.
2. The bidding credits left available for the student is set to 0.
3. Confirmation receipt of bid is printed out.

Main Success Scenario:

1. The student goes to Elective Bidding module within the student portal.
2. The list of available electives is fetched from the database by the system.
3. The student picks electives and allocates bidding credits to each one (the total amount is 100 credits).
4. The student submits the form with electives chosen.
5. The system invokes UC-02: Validate Prerequisites and checks whether all academic prerequisites are met.
6. The system checks whether any chosen elective courses have timetable conflicts.
7. The bid selection and credit allocation are committed to the database.
8. The confirmation message along with receipt is presented to the student.

9. The use case ends successfully.

Alternate Flows:

5a. Missing Prerequisite Course (Prerequisite Validation Failure)

- **5a1.** The system indicates an error specifying which specific prerequisite course is not met.
- **5a2.** The student is required to change/remove the invalid elective course.
- **5a3.** The student revises the selection of the courses and submits again (returns to step 4).

6a. Timetable Schedule Conflict («extend» UC-03)

- **6a1.** The system identifies the overlapping of the time for the two or more selected lectures/lab sessions.
- **6a2.** The system alerts the student about conflicting courses and asks to resolve the clash.
- **6a3.** The student revises the elective courses to remove timetable conflict and submits again (returns to step 4).